KLE Technological University | Creating Value, Leveraging Knowledge

Department of Computer Science and Engineering (AI)

Laboratory Plan

FMTH0303-3.1

Semester: V Year: 2026-27

Laboratory Title: Deep Learning Lab	Lab. Code: 26ECAP301
Total Hours: 42	Duration of Exam: 3hrs
Total Exam Marks: 20	Total ISA. Marks: 80
Lab. Plan Author: Umadevi F M, Sneha Varur	Date: 12-08-2026
Checked By: Dr. Uday Kulkarni	Date: 16-09-2026

Preamble:

Deep learning has emerged as one of the most transformative branches of artificial intelligence, enabling machines to learn hierarchical representations directly from data rather than relying on hand-crafted features. Its architectures form the backbone of modern applications such as image recognition, natural language understanding, recommendation systems, and generative AI. The objective of this laboratory is to develop the capability to implement deep learning architectures from foundational principles, apply them to realistic problems, and evaluate their behaviour with respect to accuracy, efficiency, and generalisation.

The lab provides students with hands-on experience across the major paradigms of deep learning — from building a neural network with backpropagation from scratch, to convolutional networks for visual understanding, recurrent networks for sequential and temporal data, transformer-based models for language, and autoencoders, generative models, and graph neural networks for learning without labels and from relational data. Each architecture has a distinct impact on real-world applications: convolutional networks drive image-based diagnosis, recurrent and transformer models power forecasting and language systems, and generative and graph-based models enable synthesis and relational reasoning. By progressing through these labs, students learn to select, implement, and evaluate the appropriate deep learning architecture for a given application, and to understand how each design choice affects a model's efficiency, accuracy, and real-world applicability.

Course Outcomes-CO

At the end of the course, students will be able to:

i. Implement fundamental neural network operations and training mechanisms using computational tools.
ii. Develop learning models for image, sequential, textual, unlabeled, generative, and relational data.
iii. Evaluate trained models through experimentation, performance comparison, and interpretation of results.
iv. Build an end-to-end deep learning solution for a real-world problem.

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Course Articulation Matrix: Mapping of Course Outcomes (CO) Program Outcomes

Laboratory (Course) Title: Deep Learning Lab Laboratory (Course) code: 26ECAP301 Semester: V Year: 2026-27

Course outcomes-CO/Program Outcomes-PO	1	2	3	4	5	6	7	8	9	10	11	PSO 1	PSO 2
1. **Implement** fundamental neural network operations and training mechanisms using computational tools.	H	M											H
2. **Develop** learning models for image, sequential, textual, unlabeled, generative, and relational data.		H	H										H
3. **Evaluate** trained models through experimentation, performance comparison, and interpretation of results.			H										H
4. **Build** an end-to-end deep learning solution for a real-world problem.		H	H					H					H

Degree of compliance **L**: Low **M**: Medium **H**: High

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Competency addressed in the Course and corresponding Performance Indicators

Competency	Performance Indicator (PI)	Course Specific Performance Indicators (CSPI)
1.4: Demonstrate competency in Computer Science and Engineering	PI 1.4.4: Apply AI/ML techniques to design and develop intelligent, scalable systems.	1.4.4.1: Apply fundamental deep learning concepts, including artificial neural networks, activation functions, loss functions, and optimization techniques, to solve computational problems.
		1.4.4.2: Design and develop deep neural network architectures appropriate for classification, regression, and pattern recognition tasks.
		1.4.4.3: Implement convolutional neural networks for feature extraction and image classification applications.
		1.4.4.4: Apply recurrent neural networks and related architectures for sequential and time-series data analysis.
		1.4.4.5: Analyse and apply appropriate regularization, normalization, optimization, and hyperparameter tuning techniques to improve deep learning model performance.
		1.4.4.6: Design and implement transfer learning-based deep learning solutions using suitable pre-trained models for real-world applications.
		1.4.4.7: Evaluate and interpret deep learning models using appropriate performance metrics and error analysis techniques.
Competency 2.1: Demonstrate an ability to identify and define complex engineering problems by understanding real-world contexts and constraints.	PI 2.1.2: Identify critical systems/sub systems, variables, and constraints that characterize the problem environment by analysing practical contexts and real-world limitations.	2.1.2.1: Identify the key components, inputs, outputs, and variables involved in a deep learning problem based on the given application context.
		2.1.2.2: Analyse datasets and problem characteristics to identify relevant features, patterns, and relationships for deep learning-based solutions.
		2.1.2.3: Identify practical constraints such as data availability, quality, computational resources, model complexity, and scalability while formulating deep learning problems.

		2.1.2.4: Analyse real-world application requirements and select appropriate deep learning approaches based on problem characteristics and constraints.
Competency 3.3: Demonstrate an ability to select an optimal design solution considering cost-effectiveness and environmental impact.	PI 3.3.2: Build models/prototypes to develop chosen design solutions.	3.3.2.1: Design and develop deep learning models/prototypes to address identified computational and real-world problems.
		3.3.2.2: Implement and train appropriate deep learning architectures using suitable datasets, frameworks, and computational resources.
		3.3.2.3: Experiment with alternative model architectures, hyperparameters, and optimization strategies to develop an effective deep learning solution.
		3.3.2.4: Analyse the performance, computational requirements, scalability, and practical feasibility of developed deep learning models/prototypes to select an appropriate design solution.
8.1: Demonstrate an ability to contribute effectively as a team member and lead diverse/multidisciplinary teams.	PI 8.1.2: Participate in team discussions, contribute ideas, provide feedback, and critique peer inputs.	8.1.2.1: Collaborate in teams to design, implement, and evaluate deep learning solutions for real-world problems.
		8.1.2.2: Participate in technical discussions, present deep learning approaches, and critically evaluate alternative model architectures proposed by team members.
		8.1.2.3: Provide constructive feedback on deep learning model design, implementation, experimental results, and performance evaluation.
PSO-2 (PO13) – Competency 1: Develop AI-based models to pre-process, analyse, and extract insights from structured and unstructured data.	PI 13.1.1: Build AI/ML models for data engineering tasks.	13.1.1.1: Pre-process and prepare structured and unstructured datasets for training deep learning models.
		13.1.1.2: Develop deep learning models using appropriate architectures for extracting meaningful features from structured, image, text, and sequential data.
		13.1.1.3: Implement and train deep learning models using suitable frameworks and computational tools.

		13.1.1.4: Apply data augmentation and feature representation techniques to improve the generalization capability of deep learning models.
PSO-2 (PO13) – Competency 1: Develop AI-based models to pre-process, analyse, and extract insights from structured and unstructured data.	PI 13.1.2: Evaluate efficiency, scalability, and accuracy of AI-based solutions.	13.1.2.1: Evaluate deep learning models using appropriate accuracy, precision, recall, F1-score, loss, and other task-specific performance measures.
		13.1.2.2: Compare alternative deep learning architectures and optimization strategies based on predictive performance and computational requirements.
		13.1.2.3: Analyse overfitting, underfitting, convergence, and generalization behaviour of deep learning models.
		13.1.2.4: Optimize deep learning models to improve computational efficiency, scalability, and predictive performance.
PSO-2 (PO13) – Competency 2: Develop AI-driven solutions to optimize, secure, and manage systems and networks.	PI 13.2.1: Develop AI-based optimized solutions for system and network management in diverse application domains (e.g., agriculture, healthcare, industry).	13.2.1.1: Apply deep learning techniques to develop intelligent solutions for domain-specific problems in areas such as healthcare, agriculture, industry, and other application domains.
		13.2.1.2: Select and adapt suitable deep learning architectures based on the characteristics and requirements of a given application domain.
		13.2.1.3: Develop and evaluate deep learning-based solutions that address real-world prediction, classification, detection, or decision-making problems.
		13.2.1.4: Analyse the practical limitations, computational requirements, and deployment considerations of deep learning solutions in real-world environments.

Eg: 1.2.3: Represents program outcome ‘1’, competency ‘2’ and performance indicator ‘3’.

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Experiment wise plan

1. **List of experiments/jobs planned to meet the requirements of the course.**

Category: Demonstration	**Total Weightage: -**		**No. of lab sessions: 2**	
Learning Outcomes: The students should be able to: 1. Demonstrate computer vision operations and filtering techniques. 2. Develop a neural network to understand its basic learning process.				
Expt./Job No.	**Experiment/job Details**	**No. of Lab. Session/s per batch (estimate)**	**Marks/ Experiment**	**Correlation of Experiment with the theory**
1	Demonstration of Computer Vision Operations and Image Filters	1	0	Image processing concepts through filtering operations
2	Design and implementation of Neural Network from Scratch	1	0	Neural network fundamentals through implementation
Category: Exercises	**Total Weightage: 40**		**No. of lab sessions: 6**	
Learning Outcomes: The students should be able to: 1. Implement fundamental deep learning architectures for image, sequence, text, generative, and relational data processing. 2. Apply appropriate deep learning models and training techniques to solve diverse computational tasks. 3. Analyze model behaviour, learned representations, training characteristics, and performance across different architectures. 4. Evaluate deep learning models for accuracy, efficiency, robustness, and suitability for deployment.				
Expt./Job No.	**Experiment/job Details**	**No. of Lab. Session/s per batch (estimate)**	**Marks/Experiment**	**Correlation of Experiment with the theory**
1	Building and Training Neural Networks from Scratch for Fundamental Learning Concepts	1	00	Neural network learning through training fundamentals

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2.	Image Classification using Convolutional and Pretrained Neural Networks for Visual Recognition Tasks	1	10	Visual recognition through convolutional transfer learning
3.	Sequence Learning using Recurrent and Temporal Neural Networks for Time-Series Prediction Tasks	1	10	Temporal learning through recurrent sequence architectures
4.	Graph Neural Networks for Relational Data Learning and Node Classification Tasks	1	10	Relational learning through graph neural networks
5.	Deep Learning Model Evaluation, Optimization and Deployment for Efficient Real-World Applications	1	10	Model evaluation, optimization and deployment practices
Category: Structured Enquiry		**Total Weightage: 20**		**No. of lab session: 2**
Learning Outcomes: The students should be able to: 1. **Investigate** the behaviour of deep learning models through controlled experimentation on a given real-world problem. 2. **Compare** alternative learning approaches using appropriate performance measures and experimental evidence. 3. **Analyse** and interpret model behaviour, strengths, limitations, and failure cases to draw evidence-based conclusions.				
1	Semantic Text Representation using Lexical and Transformer-Based Approaches	1	10	Semantic text representation, contextual embeddings, similarity-based retrieval, model comparison and interpretation
2	Unsupervised Representation Learning using Autoencoders and Self-Supervised Learning	1	10	Unlabeled data representation, autoencoder architecture,

				latent-space representation, reconstruction-based learning, self-supervised learning, representation analysis and interpretation.

Category: Course project	**Total Weightage: 20**	**No. of lab session: 4**

Learning Outcomes:

The students should be able to:

1. **Design** an end-to-end deep learning solution by selecting suitable architectures for the identified problem.
2. **Develop** and integrate the required deep learning components into a functional application.
3. **Evaluate** the developed application using appropriate performance metrics and deployment considerations.

Expt./Job No.	**Experiment/job Details**	**No. of Lab. Slots per batch (estimate)**	**Marks/ Experiment**	**Correlation of Experiment with the theory**
1.	Design and Develop an End-to-End Deep Learning Application using Appropriate Deep Neural Architectures	4	20	End-to-end application development through architecture selection

1. Materials and Resources Required:

a) Text Book
   a. Ian Goodfellow and Yoshua Bengio and Aaron Courville: Deep Learning, MIT Press, 2016.

b) Reference Book
   a. Luca Pietro Giovanni Antiga, Thomas Viehmann, Eli Stevens, Deep Learning with PyTorch Manning Publications, 2020.

c) Deep Learning Lab Manual.

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Course Project Evaluation Rubrics

Parameter	Good (4M – 5M)	Average (2M-3M)	Poor (0M – 1M)	BL	CO	PI
Problem statement identification (Review-1)	The problem statement and domain knowledge are well understood, and an exhaustive literature survey related to the problem statement has been performed.	The problem statement is identified, with poor domain understanding and a proper literature survey not carried out.	The problem statement is identified, and there is no domain understanding or a literature survey.	3	2	1.4.4 2.1.2
Parameter	**Good (4M – 5M)**	**Average (2M-3M)**	**Poor (0M – 1M)**	**BL**	**CO**	**PI**
Assessing the Implementation (Review-2)	Complete implementation of the course project along with accurate model performance results.	80% - 90% implementation of the course project along with moderate model performance results.	60%-70% implementation of the course project with poor model performance results.	4	4	3.3.2 13.1.1
Parameter	**Good (4M – 5M)**	**Average (2M-3M)**	**Poor (0M – 1M)**	**BL**	**CO**	**PI**
The demonstration of the course project (Review-3)	Model is completely working for all the test cases and with good training and testing performance.	The model is working for a few test cases and with moderate training and testing performance.	Model is not working for test cases and with poor training and testing performance.	3	5	13.1.2

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2. **Evaluation:**

Students Assessment through ISA (80%) + ESA (20%)

In Semester Assessment (80%)	Assessment	Weightage in Marks
	Exercises	40
	Structured Enquiry	20
	Course Project	20
End Semester Assessment (20%)	ESA Lab exam	**20**
	Total	**100**

Date: 16-09-2026

HOD CSE (AI)

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